

MANAV GARG

AI / ML Engineer · MLOps · End-to-End ML Systems

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PROFESSIONAL SUMMARY

AI/ML Engineer with end-to-end experience across 6 companies — from model research to production inference. Cut a 36-hour pipeline to under 5 hours, reduced LLM hallucinations by 70%+ without fine-tuning, and trimmed AWS costs by 20–25%.

EXPERIENCE

Independent Consultant — AI/ML · Vault22 · Remote

Oct 2025 – Present

- Cut infrastructure and ML pipeline costs by 20–25% by restructuring 3+ compute-heavy workloads across AWS.
- Overhauled ETL pipelines and data lake schema across 4+ data sources, eliminating recurring ingestion failures.
- Shipped an internal admin portal with Amazon QuickSight dashboards tracking 10+ KPIs for leadership and investors.

Associate Data Scientist · IHX Pvt. Ltd (A Perfios Company) · Bangalore, Onsite

Jan 2025 – Mar 2026

- Slashed data processing time by 83% — from 36 hours to under 5 — by overhauling preprocessing pipelines across 2 production datasets.
- Reduced LLM hallucination rates by 70%+ via systematic prompt engineering, achieving structured JSON outputs with zero fine-tuning.
- Deployed asynchronous ML inference systems serving 3+ concurrent production models under strict latency constraints.
- Systematized financial document extraction across 5+ document types; packaged the solution into 2 Python libraries adopted by 4 internal teams.

Industrial Trainee — AI/ML · Intel Corporation · Chennai, Onsite

May 2024 – Aug 2024

- Built an end-to-end VLM pipeline classifying 10+ contract clause types, flagging template deviations and fraud signals for legal compliance review.

Full Stack Developer & ML Engineer · Tisac Pvt. Ltd. · Remote

May 2024 – Jun 2024

- Streamlined Docker container provisioning for model training — cut environment setup from ~30 minutes to under 2 minutes and hardened the infrastructure with per-container TLS/SSL and Firebase Auth.

Full Stack Developer Intern · Hogarth India (WPP) · Gurugram, Onsite

Aug 2023 – Oct 2023

- Delivered live vote logging, code redemption, and real-time polling for Coca-Cola's Thums Up ICC World Cup 2023 campaign — 10,000+ peak user interactions.

Android Developer & ML Intern · Wipro · Chennai, Onsite

Aug 2023 – Nov 2023

- Developed a machine vision Android app classifying 5+ pathway obstruction types for Wipro rover navigation, enabling real-time clearance logging.

TECHNICAL SKILLS

ML & AI: PyTorch, TensorFlow, Scikit-Learn, OpenCV, HuggingFace Transformers, ViT, YOLO, Donut, RAG, LLM Prompt Engineering, Multimodal Architectures, Pandas, NumPy, Matplotlib

MLOps & Infrastructure: Docker, AWS (EC2, S3, SageMaker, QuickSight), Async Inference, CI/CD, ETL Pipeline Design, Data Lake Architecture, Model Serving & Monitoring, MongoDB, SQL

Languages & Frameworks: Python, C/C++, JavaScript / TypeScript, Java, Bash, HTML5/CSS3, FastAPI, Django, Flask, Node.js / Express, React / Next.js, Firebase

Tools: Git / GitHub, Linux / Unix, VS Code, Vim, Figma, Adobe Photoshop / Illustrator

PROJECTS

LuminSkin: Skin Cancer Classification using Vision Transformers | *Python, PyTorch, ViT, DeepViT* | Oct–Nov 2024

- Benchmarked ViT vs. DeepViT on a 10,000+ image skin cancer dataset, surpassing baseline CNN accuracy by 4.2% and advancing early-detection tooling.

Satellite Image Analysis using Vision Transformers | *Python, PyTorch, ViT, CCT* | Apr–May 2024

- Compared ViT vs. CCT on land-use classification across 21 categories, achieving 94.8% top-1 accuracy — enabling satellite-based environmental monitoring at scale.

KeyKeg — Password Strength Analyzer & Manager | *Python, Scikit-Learn, Random Forest, Next.js* | Apr 2024

- Assembled an ML-powered password manager with a Random Forest classifier trained on 500K+ entries, achieving 96% strength-classification accuracy with secure vault storage.

NSCHR — Web HMIS Portal | *Express, EJS, MongoDB, CSS3* | Sep 2021

- Conceived and sold a Health Management Information System to 2 local clinics, cutting patient record retrieval time from ~5 minutes to under 30 seconds at point of care.

Snippy — AI Short-Form Video Generator | *Python, Flask, MoviePy, Whisper, ElevenLabs* | 2024

- Engineered an AI video pipeline (monologue/dialogue) integrating Edge TTS, Whisper transcription, GPT/Gemini script generation, and context-aware image overlays via a Flask UI.

Rekol — WhatsApp & Web Productivity Assistant | *Node.js, MongoDB, WhatsApp Web API* | 2023

- Developed a WhatsApp-native assistant for reminders, tasks, and notes — with a web dashboard and email notifications backed by MongoDB.

EDUCATION

B.Tech, Computer Science Engineering

Graduated Jun 2025

Vellore Institute of Technology, Chennai · *Specialization: AI & ML* · *CGPA: 8.51 / 10*

Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, XAI, Linear Algebra, Probability & Statistics, OOP